Girijananda Chowdhury University, Guwahati, 781017, Assam

Formative Assessment – I

5th Semester, B.Tech CE (2026-27) | **Structural Design-I (BCE23501T)**

Date: 15-10-2026 | **Batch: 2024-28**

Duration: 1 Hour | **Total Marks: 20**

1. Answer the following: **1+1+1+1= 4**

(i) The maximum strain in the tension reinforcements in a rectangular section at the time of failure should not be less than_____. **(CO1)**

(ii) Neutral axis depth factor for a balanced section, in WSM, can be expressed as_____. **(CO1)**

(iii) Multiple safety factors are used in **(CO1)**

(a) Working stress method, (b) Ultimate load method,

(c) Limit state method, (d) None of the above.

(iv) Write the empirical expression for modulus of elasticity of concrete as recommended by IS 456:2000. **(CO1)**

2. A singly reinforced RCC section has dimensions 300 mm×650 mm. 4 nos. of 25 mm dia. Fe415 bars are provided at an effective cover of 50 mm. Grade of concrete used is M20. Using LSM, find: (a) Actual depth of neutral axis, (b) Critical depth of neutral axis and (c) Moment of resistance of the given section, **3+3+2 = 8 (CO2)**

3. Find the depth and area of steel required in a simply supported rectangular RCC beam of width 350 mm, subjected to a live load of 30 kN/m over an effective span of 8 m. Use M20 grade of concrete and Fe415 grade of steel. Use any other data, if required. **8 (CO2)**